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Project Report:

**Prediction of Potential Wildfire Regions Using
Multi-Modal Analysis and Drone-Captured Data**

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Dedicated to the brave firefighters and communities affected by wildfires worldwide.

Abstract

This report details the development of a comprehensive, multi-modal machine learning framework for proactive wildfire prediction and management. Leveraging diverse data streams, primarily from drone-captured imagery and meteorological sensors, the project addresses the critical need for early detection, dynamic behavior prediction, and optimized resource allocation. The methodology encompasses advanced data preprocessing, visual semantic segmentation for fuel mapping, multi-modal visual classification for localized fire detection, and sophisticated meteorological classification and regression models for fire danger rating and severity prediction. A key innovation includes the development of intelligent synthetic ground truth heatmaps for training, addressing data scarcity in real-world scenarios. The report outlines the architecture, implementation steps, statistical analyses, and performance evaluation of each model, culminating in a proposed multi-model operational framework that integrates these components for a unified, impactful wildfire prediction system. The findings demonstrate the potential for significantly enhancing situational awareness and supporting more effective, proactive responses to wildfire threats.

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Chapter 1

Problem Statement

1.1 Background and Motivation

Wildfires represent an increasingly severe global crisis, causing extensive damage to ecosystems, economies, and human lives. The destruction of vast forest and grassland areas, displacement of communities, pervasive air pollution, and acceleration of climate change underscore the urgent need for more effective wildfire management strategies. Current approaches, which often rely on infrequent satellite imagery, ground patrols, and human observation, are frequently hampered by limitations in real-time capability, spatial coverage, and inherent dangers, especially in remote and inaccessible terrains.

1.2 Project Vision and Scope

This project outlines a comprehensive vision for a full-fledged wildfire prediction system using heat map analysis and image recognition, incorporating drone-based deployment for analysis over large and hard-to-access areas. The scope of the immediate project implementation focuses on a primitive analysis using readily available datasets to validate core methodologies and demonstrate proof-of-concept before scaling to the full complexity of real-time drone data integration. The project aims to transition wildfire management from a reactive approach to a proactive, predictive one.

1.3 Improvements and Innovations

This project introduces several key novelties and contributions:

- a) **Intelligent Synthetic Ground Truth Heatmap Generation:** Addressing the critical challenge of scarce pixel-level pre-ignition ground truth data, a novel method for generating synthetic heatmaps has been developed. This method leverages the affinity of actual high-temperature zones (detected via red and green channel spikes in thermal imagery) to create realistic training labels for wildfire potential, providing a meaningful direction for model learning without real, large-scale ground truth.
- b) **Comprehensive Multi-Modal Fusion for Prediction:** Integration of diverse data types including RGB imagery, thermal imagery, and extensive meteorological/soil sensor data in a unified predictive framework.
- c) **Advanced Feature Engineering for Time-Series Data:** Development of sophisticated Fire Weather Index (FWI)-like features and extensive lagged/rolling window statistics tailored for chronological meteorological data, enhancing predictive power.
- d) **Multi-Model Ensemble for Robustness:** Utilization of multiple specialized machine learning models (semantic segmentation, visual classification, meteorological classification, meteorological regression) to address different facets of wildfire prediction, ensuring robustness and adaptability to various operational scenarios.
- e) **Drone-Centric Data Acquisition:**** The entire framework is designed to seamlessly integrate with drone-captured data, enabling rapid, localized, and on-demand data collection in remote

or dangerous areas, which is a significant advancement over traditional static sensor networks or infrequent satellite passes.

Chapter 2

Data Acquisition and Preprocessing

2.1 Data Sources

This section details the various data sources utilized throughout the project, encompassing both publicly available datasets and conceptual data types derived from drone operations.

2.1.1 Hourly Weather Dataset

This dataset, referenced from [8], provides hourly meteorological and soil conditions relevant for wildfire prediction. It comprises:

- **Temporal Features:** Date and time stamps.
- **Meteorological Conditions:** Temperature (2m), relative humidity (2m), precipitation, surface pressure, cloud cover, wind speed (10m).
- **Soil Conditions:** Soil temperature (0 – 7cm and 7 – 28cm depths) and soil moisture (0 – 7cm and 7 – 28cm depths).
- **Target Variables:** A binary flag ('forest_fire': 'Y'/'N') indicating fire occurrence, and a continuous 'Severity' metric.

This dataset was foundational for the Meteorological Classification and Meteorological Regression models.

2.1.2 FLAME 3 Computer Vision Dataset

The FLAME 3 dataset [3] is a collection of aerial Unmanned Aerial Vehicle (UAV)-collected side-by-side multi-spectral wildfire imagery designed for computer vision tasks. Each entry is an image quartet:

- **Raw RGB Image:** 4000×3000 pixels, .JPG format.
- **Raw Thermal Image:** 640×512 pixels, .JPG format.
- **Corrected FOV RGB Image:** 640×512 pixels, .JPG format (spatially aligned with thermal).
- **Thermal TIFF:** 640×512 pixels, .TIFF format, containing per-pixel Celsius temperature values.

This dataset was foundational for the Image Preprocessing, Visual Classification (Heatmap Prediction), and Visual Semantic Segmentation components.

2.1.3 Forest Cover Type Dataset

The Forest Cover Type Dataset [9] from the UCI Machine Learning Repository provides a different type of environmental data, focusing on forest characteristics which can be relevant for fuel mapping and geospatial prediction. It consists of various cartographic variables for predicting forest cover type.

2.1.4 FLAME Dataset (IEEE DataPort)

This refers to a general FLAME dataset hosted on IEEE DataPort [10], providing aerial imagery for pile burn detection using drones. This resource serves as a broader collection of FLAME series data beyond the specific FLAME 3 version used for detailed computer vision tasks.

2.1.5 Custom Annotated Segmentation Dataset

For the Visual Semantic Segmentation task, a dataset consisting of RGB images ('image_<number>.jpg') and corresponding pixel-level binary masks ('image_<number>.png') was utilized. This dataset, conceptually similar to [4], required specific preprocessing for efficient model training.

2.1.6 General Wildfire Data Sources

Additional reference data for contextual understanding and external validation were drawn from established wildfire monitoring sources:

- **FIRMS NASA Earthdata** [1], **LANCE FIRMS Active Fire Data** [2]: For active fire detection and monitoring.
- **2010-SOUTH TRAIL Wildfire Boundary Dataset** [5]: Digital aerial imagery (RGB and multispectral) related to historical fire boundaries.
- **U.S. Geological Survey (USGS) Wildland Fire Science** [6]: General wildland fire research and data.
- **National Interagency Fire Center (NIFC)** [7]: National fire information and statistics.

2.2 Image Preprocessing and Standardization (Image_Preprocessing_Standardized.ipynb)

This initial and critical phase transforms raw, heterogeneous image data into a standardized, consistent, and high-quality format, indispensable for the effective training and reliable operation of subsequent machine learning models.

- a) **Pairing RGB and Thermal Images:** Images (and hypothetical videos) are paired based on their common filename identifier (e.g., '00001' for '00001_aligned_rgb.jpg' and '00001_regenerated_thermal.jpg'). For the FLAME 3 dataset, this pairing is inherently supported by its structure.
- b) **Extraction and Regeneration of Thermal Data:**
 - While the FLAME 3 dataset provides a Thermal TIFF with radiometric Celsius values, the pipeline focuses on processing the '_regenerated_thermal.jpg' files.
 - The thermal JPG is processed to enhance and standardize thermal representation. This involves loading the color thermal JPG, extracting the combined intensity of its red and green channels (as a proxy for heat spikes), and resizing it to a consistent dimension (e.g., 640×512).
- c) **Alignment of RGB Images:** RGB images are scaled, cropped, and translated to align spatially with the thermal images. The process prioritizes using the 'Corrected FOV RGB Image' (which is pre-aligned) and falls back to performing a resize and center-crop on the 'Raw RGB Image' if the corrected version is unavailable. All output images are resized to 640×512 pixels.
- d) **Output Generation:** The paired, aligned, and regenerated images (e.g., '*_aligned_rgb.jpg', '*_regenerated_thermal.png') are copied into structured output folders ('processed_flame3_dataset/Fire' and 'No_Fire'), ensuring consistency for downstream models.

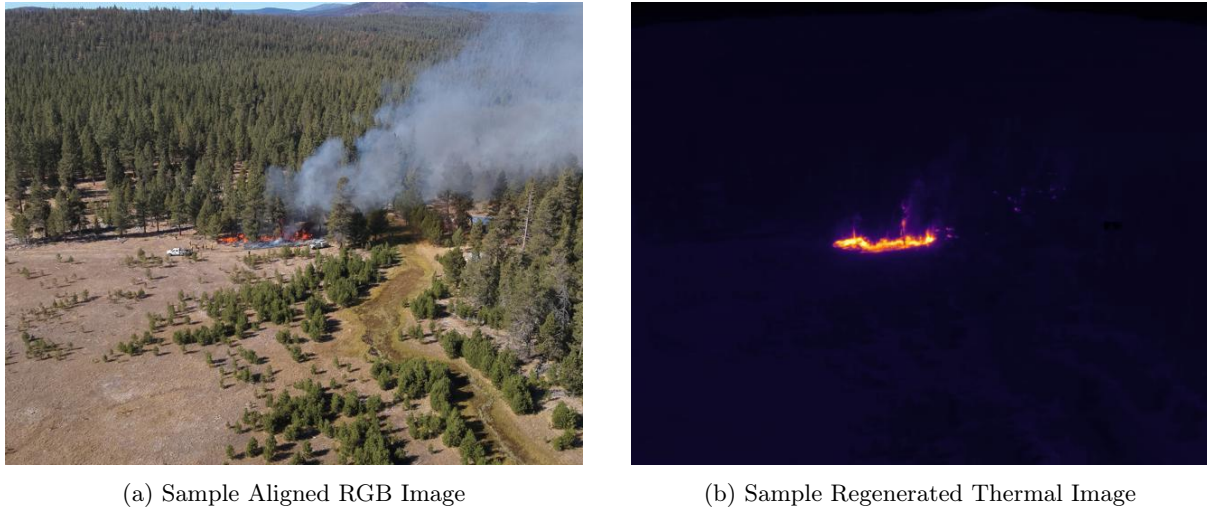


Figure 2.1: Output of Image Preprocessing and Standardization Pipeline. Images are resized to a uniform 640×512 resolution.

2.3 Meteorological Data Preprocessing and Feature Engineering

This stage transforms raw hourly weather data into a rich feature set suitable for machine learning.

- a) **Chronological Sorting:** The dataset is sorted strictly by 'datetime' to ensure the integrity of lagged and rolling window features, which rely on temporal order. This also prevents future data leakage during train/test splitting.
- b) **Temporal Feature Extraction:** 'hour_of_day', 'day_of_week', 'month', 'day_of_year', and 'week_of_year' are extracted to capture diurnal and seasonal patterns in fire risk.
- c) **Target Encoding:** For classification, the 'forest_fire' column ('N'/'Y') is encoded to '0'/'1'. For regression, 'Severity' is retained as a numerical target.
- d) **Feature Exclusion for Data Leakage:** The 'Severity' column, despite its variability, was identified as having an extremely high correlation (≈ 0.995) with the 'forest_fire' classification target. This indicated data leakage, as 'Severity' is likely a consequence of fire occurrence rather than a predictor. Consequently, 'Severity' was definitively dropped from the feature set for the classification task to ensure a realistic model.
- e) **Comprehensive Feature Engineering (FWI-like Lag/Rolling):**
 - **FWI-like Indices:** Derived features such as 'dryness_index' (temperature and humidity interaction), 'wind_dryness_factor' (wind and dryness interaction), 'avg_soil_moisture', and 'avg_soil_temp' are created to consolidate environmental conditions into more meaningful fire risk indicators.
 - **Diverse Lag Features:** Values from previous hours (1, 3, 6, 12, 24, 48 hours ago) for key meteorological variables ('temperature_2m', 'relative_humidity_2m', 'wind_speed_10m', 'precipitation', 'dryness_index') are included. This captures immediate and short-term historical context.
 - **Rolling Window Features:** Aggregated statistics (mean, sum, max) over various time windows (3, 6, 12, 24, 48 hours) are computed for relevant features (e.g., 'temp_24h_avg', 'humidity_48h_avg', 'precip_6h_sum'). These features capture longer-term trends and cumulative effects.
- f) **NaN Handling:** Rows containing 'NaN' values introduced by lagging/rolling operations (typically at the beginning of the DataFrame) are explicitly dropped. Any remaining 'NaN' values from original data are imputed (e.g., mean for numerical, 0 for precipitation sums).
- g) **Preprocessing Pipeline (StandardScaler OneHotEncoder):** A 'ColumnTransformer' pipeline scales numerical features using 'StandardScaler' and one-hot encodes categorical features.
- h) **Chronological Train/Test Split:** The dataset is split into training (80%) and testing (20%) sets based on chronological order. This is vital for time-series data to prevent future data leakage and ensure that the model is evaluated on truly unseen future conditions.

2.4 Statistical Analysis of Meteorological Data

Statistical analysis was performed to understand feature distributions, inter-feature relationships, and feature-target correlations, informing model selection and confirming data integrity.

a) **Target Variable Distribution (Severity):**

- **Observation:** The ‘Severity’ target variable exhibited a highly skewed distribution, with a significant majority of values concentrated at 0 (mean: 6.26, max: 42.64, 25/50/75% quartiles at 0). This directly reflects the real-world rarity of significant wildfire events.
- **Significance:** This observation was crucial for immediately ruling out standard linear regression models due to their assumptions about target distribution and for favoring robust tree-based models capable of handling zero-inflated and highly skewed data.

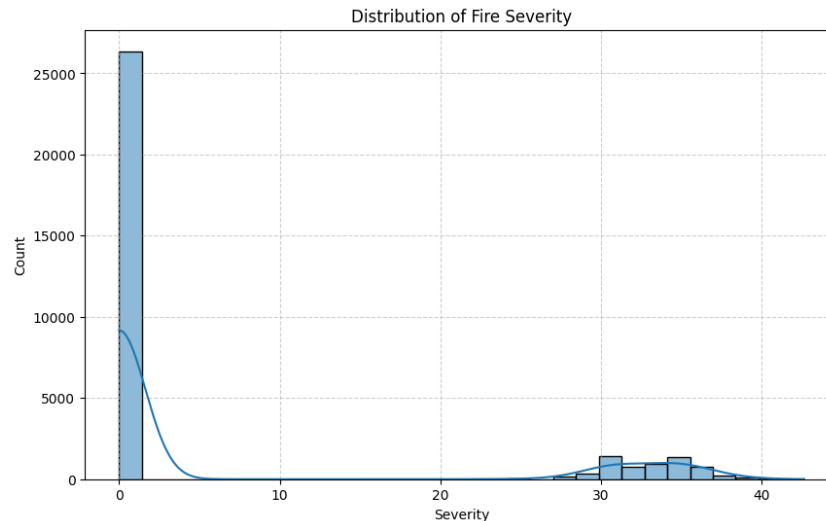


Figure 2.2: Distribution of Fire Severity in Training Data. Note the high concentration at 0.

b) **Feature Distributions (Histograms and Box Plots):**

- **Observation:** Histograms revealed diverse distributions for numerical features (e.g., skewed for precipitation, more normal-like for temperature). Box plots showed potential outliers in features like wind speed and precipitation.
- **Significance:** Confirmed the need for models robust to various distributions and outliers (e.g., tree-based ensembles) and validated the use of ‘StandardScaler’ for preprocessing.

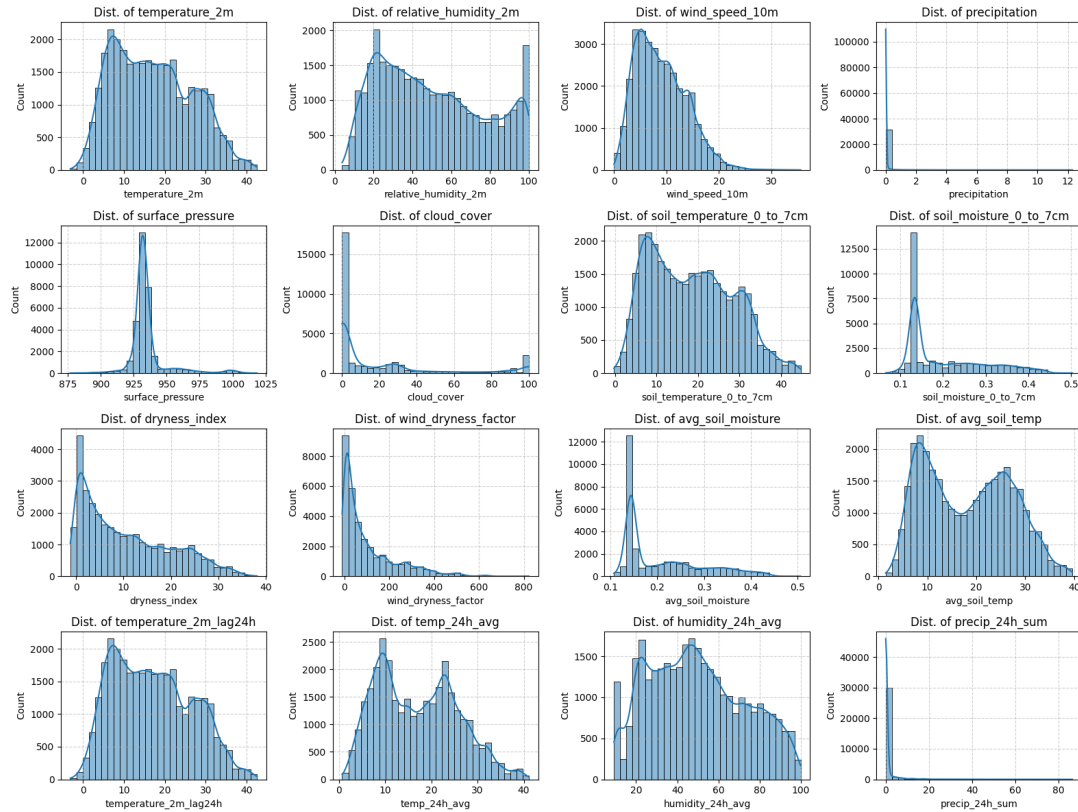


Figure 2.3: Sample Histograms of Key Numerical Features (e.g., temperature, humidity, dryness index).

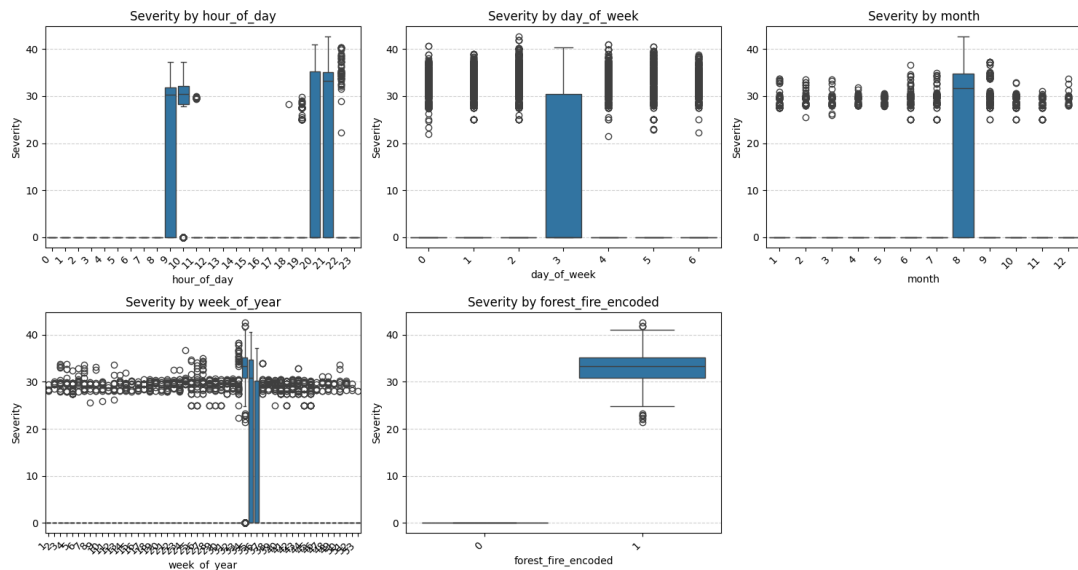


Figure 2.4: Sample Box Plots of Key Numerical Features (e.g., temperature, humidity, showing outliers).

c) Correlation Analysis (Feature-Target Feature-Feature):

- **Observation (Initial - Classification Notebook):** A high absolute correlation of ≈ 0.995 between 'Severity' and 'forest_fire' was initially observed.
- **Significance (Data Leakage Elimination):** This finding was critical in identifying and eliminating data leakage. 'Severity' was dropped from features in the classification task, as it was deemed a consequence of fire (rather than a causal predictor) and using it would lead to unrealistic performance.
- **Observation (Post-Leakage Removal New Features):** After dropping 'Severity' and adding new features, the strongest feature-target correlations observed were:

‘soil_temperature_7_to_28cm’ (0.516), ‘temp_48h_avg’ (0.493), ‘humidity_48h_avg’ (-0.426), ‘relative_humidity_2m_lag48h’ (-0.384), and engineered indices like ‘dryness_index’ (0.425).

- **Significance:** These correlations confirm the expected physical relationships (e.g., higher temperatures/dryness increase fire risk/severity, higher humidity/moisture decrease it). They validate that the engineered FWI-like and lagged/rolling features effectively capture predictive information without introducing leakage, supporting their inclusion in the models.

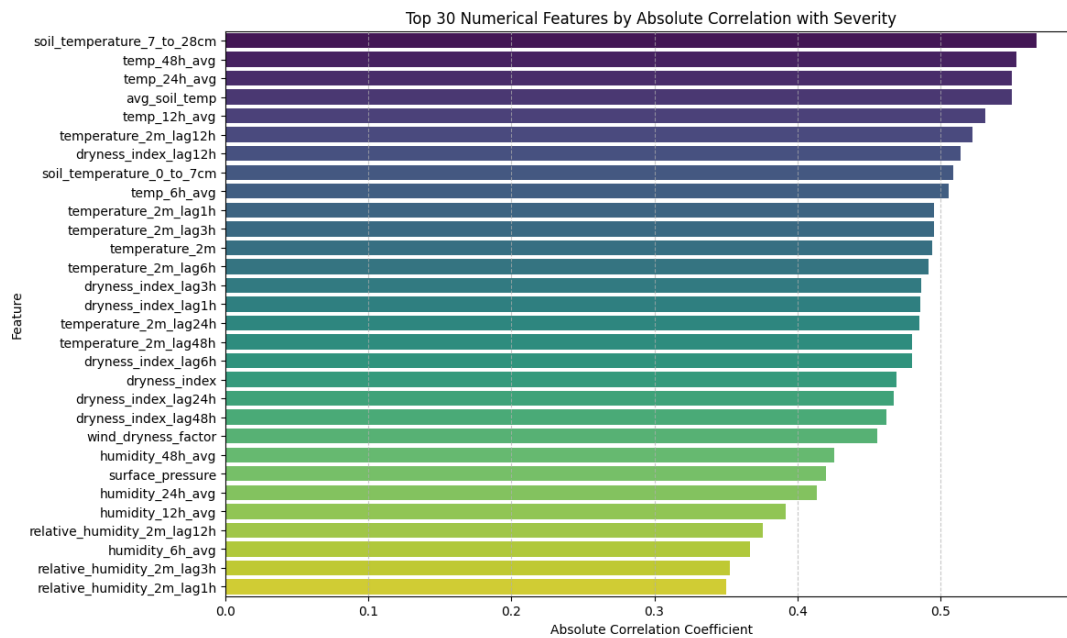


Figure 2.5: Top 30 Numerical Features by Absolute Correlation with Severity.

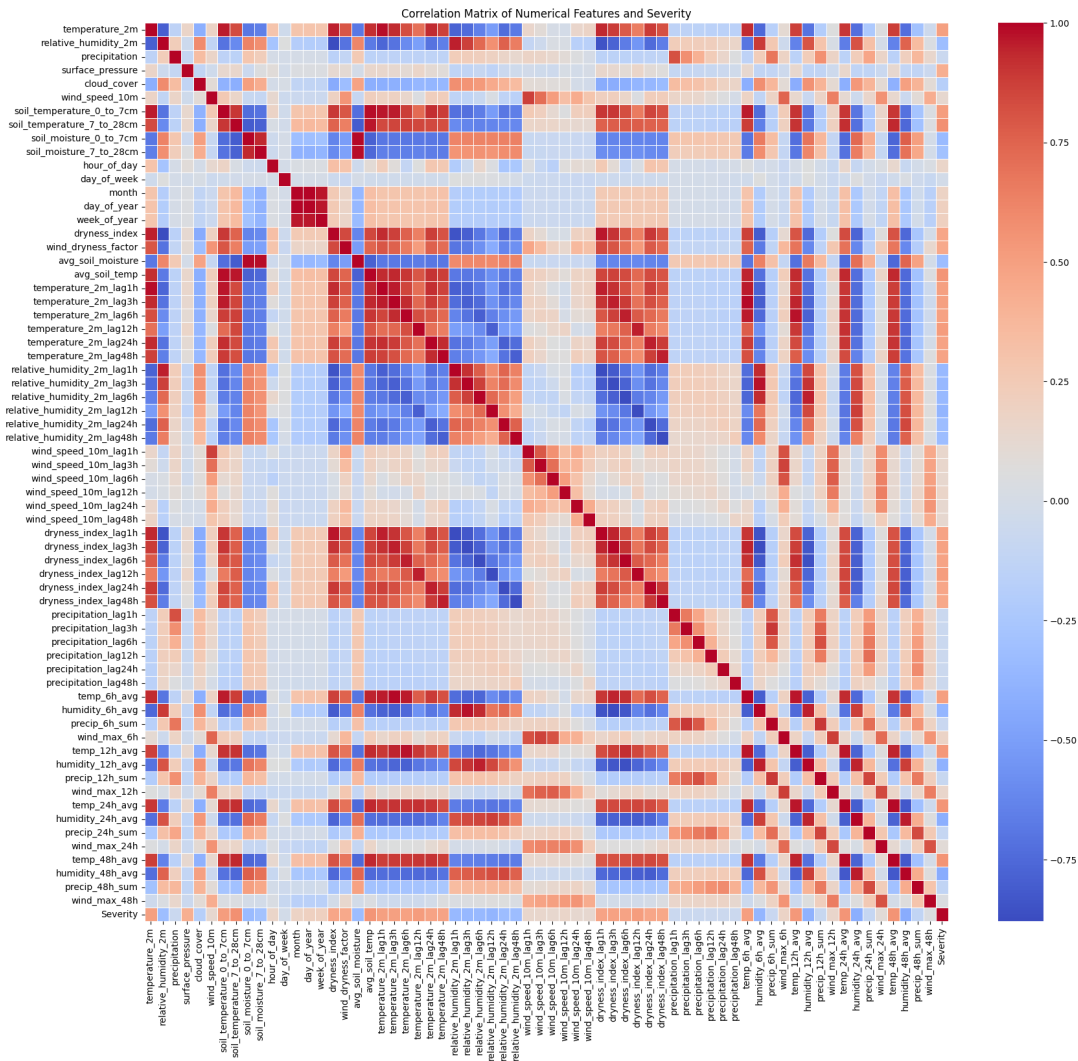


Figure 2.6: Full Correlation Matrix of Numerical Features (X_{train}) for Regression.

d) **Relationship between Categorical Features and Target (Box Plots):**

- **Observation:** Box plots of 'Severity' across 'hour_of_day', 'month', 'day_of_week', and 'forest_fire_encoded' (as an input feature for regression) showed varying distributions, indicating that Severity is higher in specific temporal windows and significantly elevated when 'forest_fire_encoded' is 1 (i.e., when a fire generally occurred).
- **Significance:** This validates the importance of these categorical and temporal features in predicting severity and justifies their one-hot encoding in the preprocessing pipeline.

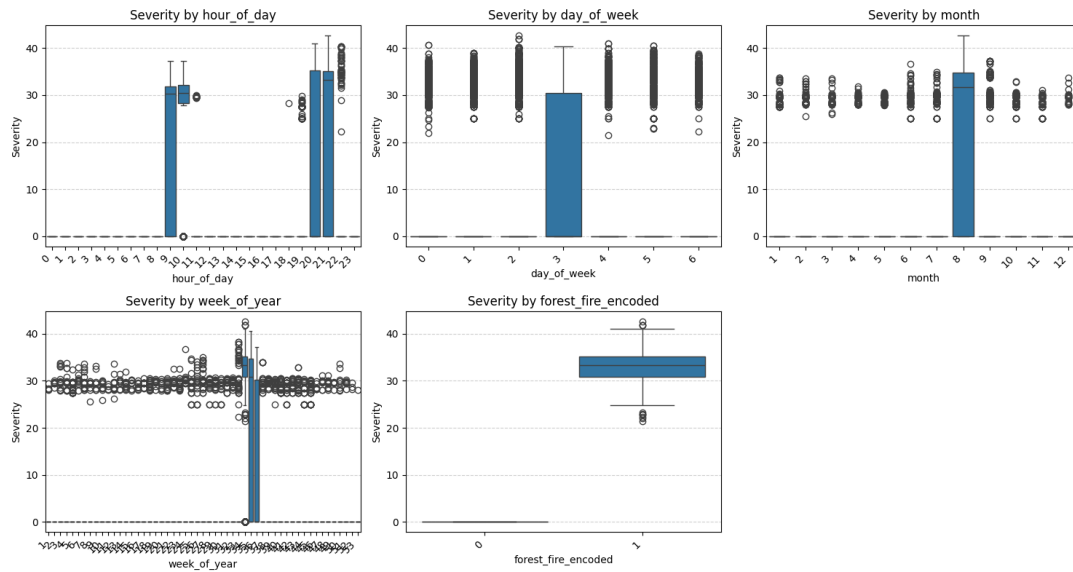


Figure 2.7: Box Plots of Severity Distribution Across Categorical Features (e.g., Hour of Day, forest_fire_encoded).

Chapter 3

Model Architectures and Methodologies

This chapter details the machine learning models developed, each tailored for specific aspects of wildfire prediction.

3.1 Visual Classification / Heatmap Prediction (Heatmap_Classification_Prediction.ipynb)

3.1.1 Purpose and Role

This model is designed for localized fire detection within drone imagery. It takes both RGB and thermal image inputs to predict the probability of fire in small, discrete grid cells across the scene, generating a heatmap that highlights potential ignition points or active fire areas.

3.1.2 Inputs and Outputs

- **Inputs:** Pairs of preprocessed RGB images ('*_aligned_rgb.jpg') and thermal images ('*_regenerated_thermal.jpg'), split into 32×32 pixel patches.
- **Outputs:** A 512×640 (image-sized) heatmap where each pixel represents the predicted probability (0-1) of fire presence in its corresponding 32×32 grid cell.

3.1.3 Model Architecture: Multi-Modal Fusion Network

The model is a convolutional neural network (CNN) featuring two parallel branches:

- **RGB Branch:** Processes $32 \times 32 \times 3$ RGB patches through convolutional layers, batch normalization, Leaky ReLU activations, and max-pooling, followed by a flattening layer.
- **Thermal Branch:** Processes $32 \times 32 \times 1$ thermal (grayscale) patches through a similar convolutional structure (potentially with fewer filters per layer initially, reflecting its single channel), also ending with a flattening layer.
- **Fusion Layer:** The flattened outputs from both branches are concatenated and fed into a series of dense layers with Leaky ReLU activations and batch normalization, culminating in a sigmoid activation function for a single probability output per patch.

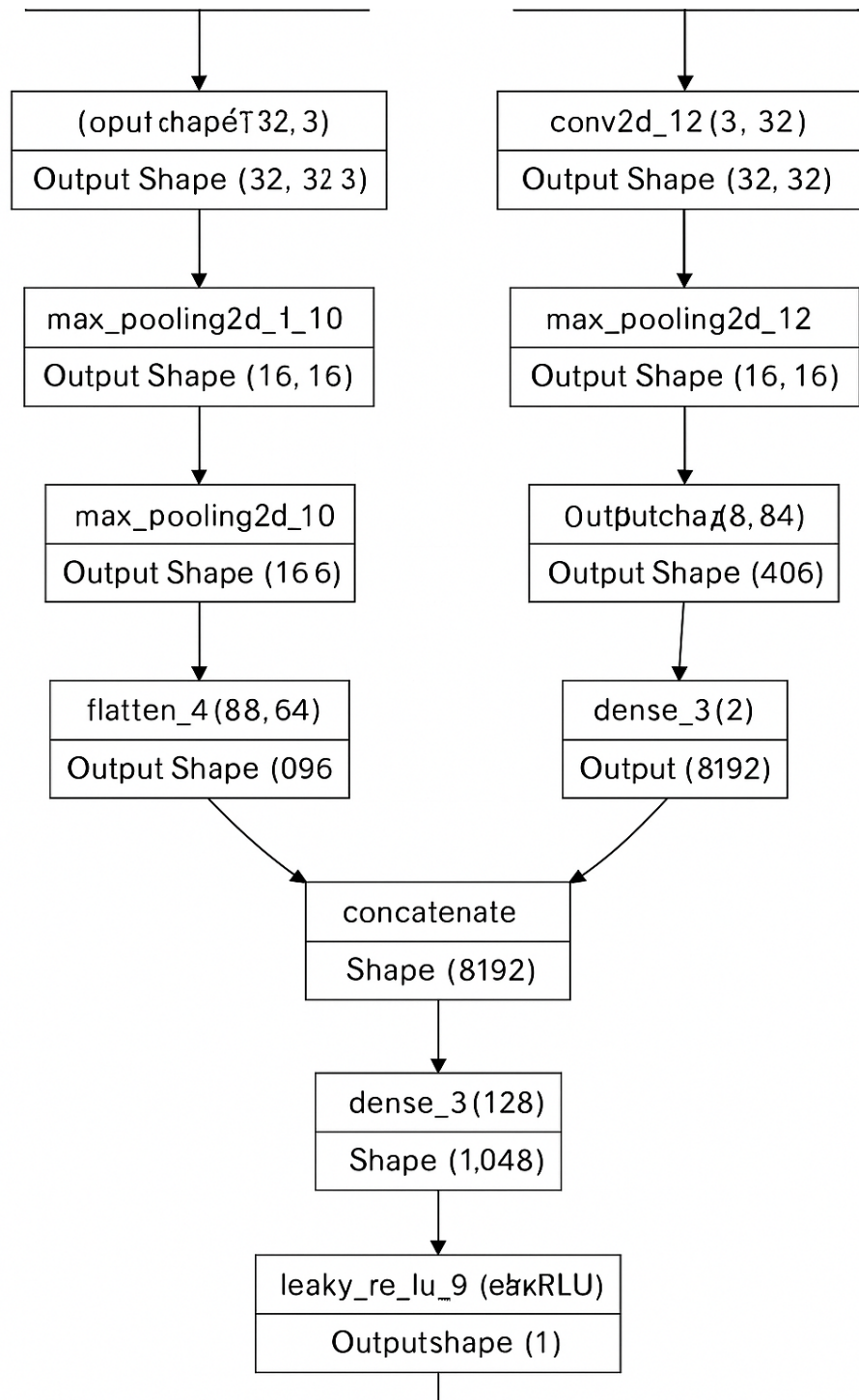


Figure 3.1: Conceptual Diagram of the Multi-Modal Fusion Network Architecture.

3.1.4 Novelty: Intelligent Synthetic Ground Truth Heatmap Generation

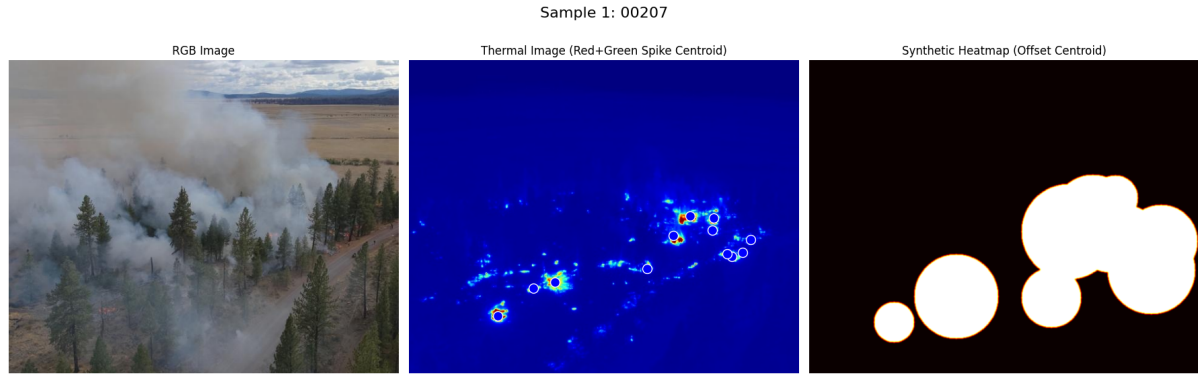
A significant challenge in training models for proactive wildfire prediction is the scarcity of actual pixel-level ground truth data for "potential fire" or "ignition probability" before a fire fully develops. To address this, a novel approach for generating synthetic ground truth heatmaps was implemented:

- For images from the 'Fire' folder: The corresponding '_regenerated_thermal.jpg' image is loaded. The red and green channels (known to be more saturated in hot regions of these specific thermal images) are analyzed. Hotspot centroids are identified by thresholding (e.g., ≥ 180) and contour detection on the combined red/green intensity. Synthetic "risk areas" (e.g., filled circles with

Gaussian blur) are then drawn on a blank heatmap image, with a small random offset (up to ± 50 pixels) from these detected thermal centroids. This provides a "direction prediction" for the model to learn relationships between visible/thermal cues and potential hotspots.

- For images from the '**No_Fire**' folder: No centroid detection or hotspot generation is performed. The synthetic ground truth heatmap is an entirely blank (all zeros) image, accurately reflecting zero fire risk.

This method allows for model training and architectural validation without relying on scarce real ground truth heatmaps, providing a valuable proxy for early-stage development.



(a) Sample Image Process Flow

Figure 3.2: Visualizing Intelligent Synthetic Heatmap Generation. For 'Fire' images, synthetic hotspots are created near actual thermal heat sources with a slight offset.

3.2 Visual Semantic Segmentation (Visual_Semantic_Segmantation.ipynb)

3.2.1 Purpose and Role

The semantic segmentation model performs pixel-level classification, allowing for precise delineation of objects and regions within drone imagery. This is crucial for detailed fuel mapping, identifying natural barriers, and post-fire burn severity assessment.

3.2.2 Inputs and Outputs

- **Inputs:** RGB images ('image_<number>.jpg') and corresponding binary masks ('image_<number>.png') from 'data/images' and 'data/masks' directories.
- **Preprocessing:** Images and masks are resized to 192×192 pixels for computational efficiency, normalized to $[0, 1]$, and augmented (flip, rotation, brightness/contrast) to increase data diversity. Masks are resized using nearest-neighbor interpolation to preserve class integrity.
- **Outputs:** A binary mask (192×192) representing the segmented regions (e.g., fire/no fire, specific vegetation types).

3.2.3 Model Architecture: U-Net

A U-Net architecture is employed, characterized by its symmetric encoder-decoder structure with skip connections:

- **Encoder (Contracting Path):** Consists of repeated convolutional blocks (Conv2D $\rightarrow$ Batch Normalization $\rightarrow$ ReLU) followed by max-pooling. This path extracts context by progressively reducing spatial dimensions and increasing feature channels.
- **Decoder (Expanding Path):** Symmetrically upsamples the features using transposed convolutions, concatenates them with high-resolution features from the encoder via skip connections, and then applies more convolutional blocks. Skip connections allow the model to leverage both high-level semantic information and fine-grained spatial details.

- **Output Layer:** A final 1×1 convolutional layer with a sigmoid activation function outputs a binary segmentation mask.

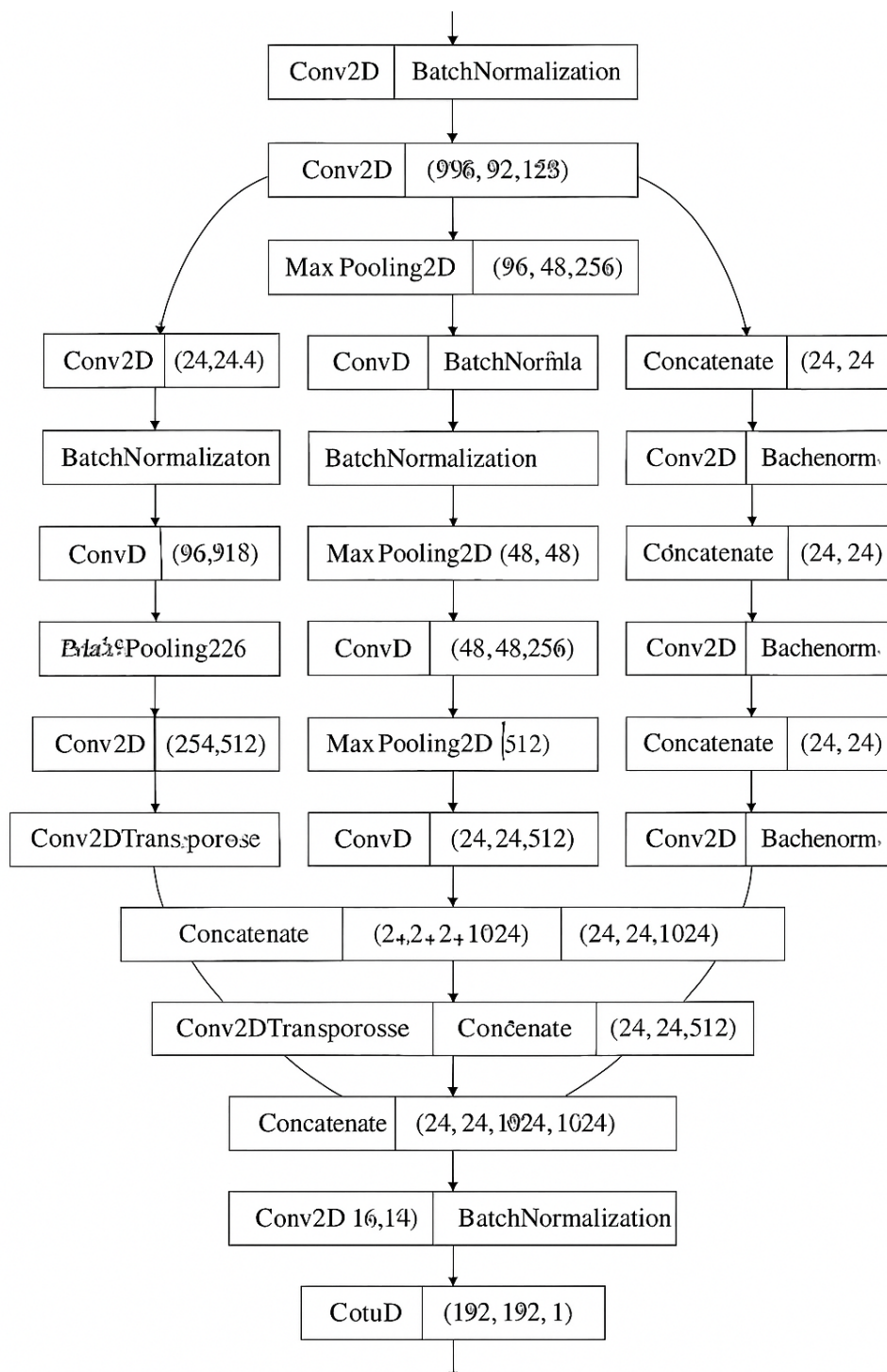


Figure 3.3: Standard U-Net Architecture for Semantic Segmentation.

3.3 Meteorological Classification (Meteorological_Classification.ipynb)

3.3.1 Purpose and Role

This model classifies hourly weather conditions into discrete fire danger categories. It provides a high-level alert system based on atmospheric and soil conditions conducive to fire ignition.

3.3.2 Inputs and Outputs

- **Inputs:** Extensive tabular features derived from ‘Hourly_Weather_Dataset.csv’, including:
 - Raw meteorological/soil readings: ‘temperature_2m’, ‘relative_humidity_2m’, ‘precipitation’, etc.
 - Temporal features: ‘hour_of_day’, ‘day_of_week’, ‘month’, ‘day_of_year’, ‘week_of_year’.
 - FWI-like indices: ‘dryness_index’, ‘wind_dryness_factor’, ‘avg_soil_moisture’, ‘avg_soil_temp’.
 - Diverse Lagged Features: ‘*_lag1h’, ‘*_lag3h’, ‘*_lag6h’, ‘*_lag12h’, ‘*_lag24h’, ‘*_lag48h’ for core numerical features.
 - Rolling Window Features: ‘*_Xh_avg’, ‘*_Xh_sum’, ‘*_Xh_max’ for various windows (e.g., 6h, 12h, 24h, 48h).
- **Output:** A binary classification (0: ‘No Fire’, 1: ‘Fire’) indicating the probability of fire occurrence.

3.3.3 Preprocessing and Feature Engineering Highlights

- **Chronological Split:** Data is strictly split chronologically (80% train, 20% test) to prevent data leakage and ensure realistic evaluation on future data.
- **Data Leakage Mitigation:** The ‘Severity’ column was definitively dropped from the feature set due to its ≈ 0.995 correlation with the ‘forest_fire’ target, confirming it as a direct consequence/leak.
- **Class Imbalance Handling (SMOTE):** Given the rarity of ‘Fire’ events, Synthetic Minority Over-sampling Technique (SMOTE) was applied to the training data to balance the classes, improving the model’s ability to learn from the minority class.

3.3.4 Models Evaluated

Three Gradient Boosting Machine (GBM) variants and a Random Forest Classifier were evaluated:

- **Random Forest Classifier:** Ensemble of decision trees (bagging).
- **XGBoost Classifier:** Gradient Boosting Machine (GBM).
- **LightGBM Classifier:** Optimized GBM.

3.4 Meteorological Regression (Severity Prediction) (Meteorological_Regression.ipynb)

3.4.1 Purpose and Role

This model provides a quantitative prediction of wildfire ‘Severity’, offering a continuous numerical estimate of fire impact or intensity based on meteorological and environmental factors. This complements the classification model by providing fine-grained severity levels.

3.4.2 Inputs and Outputs

- **Inputs:** The same extensive tabular feature set as the Meteorological Classification model. The ‘forest_fire_encoded’ (0/1) feature is now treated as a categorical input feature to predict ‘Severity’.
- **Output:** A continuous numerical value representing the predicted ‘Severity’. Negative predictions are clipped to 0.

3.4.3 Preprocessing Highlights

- **Chronological Split:** Similar to classification, a chronological split (80% train, 20% test) is employed.
- **Target Distribution Handling:** The ‘Severity’ target is highly skewed with many zeros. Tree-based models are inherently more robust to this.

- **Comprehensive Feature Set:** Includes all engineered temporal, FWI-like, and lagged/rolling features, with ‘forest_fire_encoded’ included as a predictor.
- **NaN Handling:** Robust initial and post-feature engineering NaN handling (ffill/bfill and dropna) ensures data integrity.

3.4.4 Models Evaluated

Three ensemble regression models were evaluated:

- **Random Forest Regressor:** Ensemble of decision trees for regression.
- **XGBoost Regressor:** Gradient Boosting Machine for regression. **LightGBM Regressor:** Optimized GBM for regression.

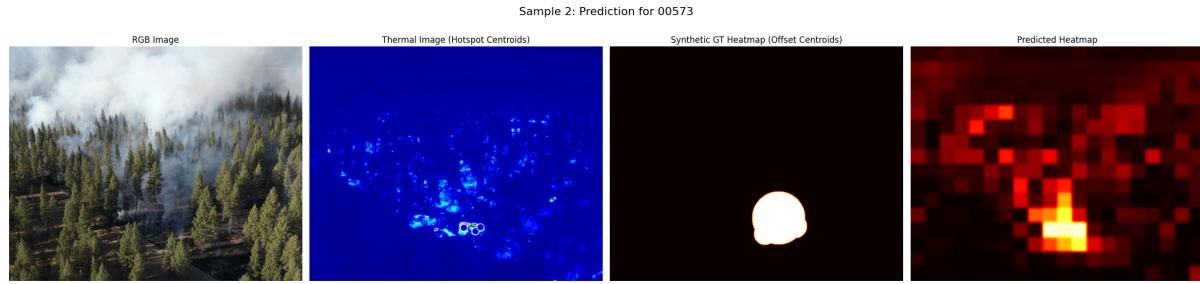
Chapter 4

Experimental Setup and Results

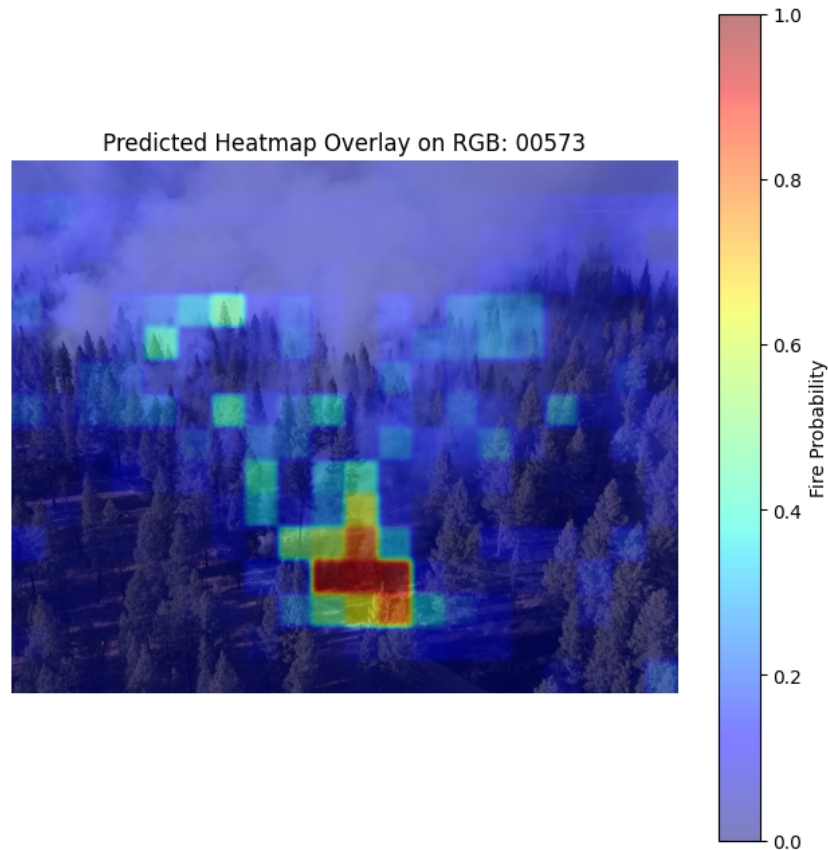
This chapter presents the key results from the training and evaluation of the developed models.

4.1 Image Preprocessing Verification

The image preprocessing pipeline successfully standardized image dimensions and formats.



(a) Sample Processed Image For Imaging Prediction



(b) Final Output For Imaging Prediction

Figure 4.1: Sample Outputs from Image Processing and Wildfire Prediction, showing consistent sizing and standardized formats for downstream models.

4.2 Visual Semantic Segmentation Results

The U-Net model was trained on the preprocessed image and mask pairs (192×192 pixels).

- **Training Objective:** To learn pixel-level mapping between environmental features in RGB images and their corresponding masks (e.g., fire pixels, fuel types).
- **Performance Metrics:** Typically evaluated using Binary Accuracy and Mean Intersection over Union (IoU) for binary masks.

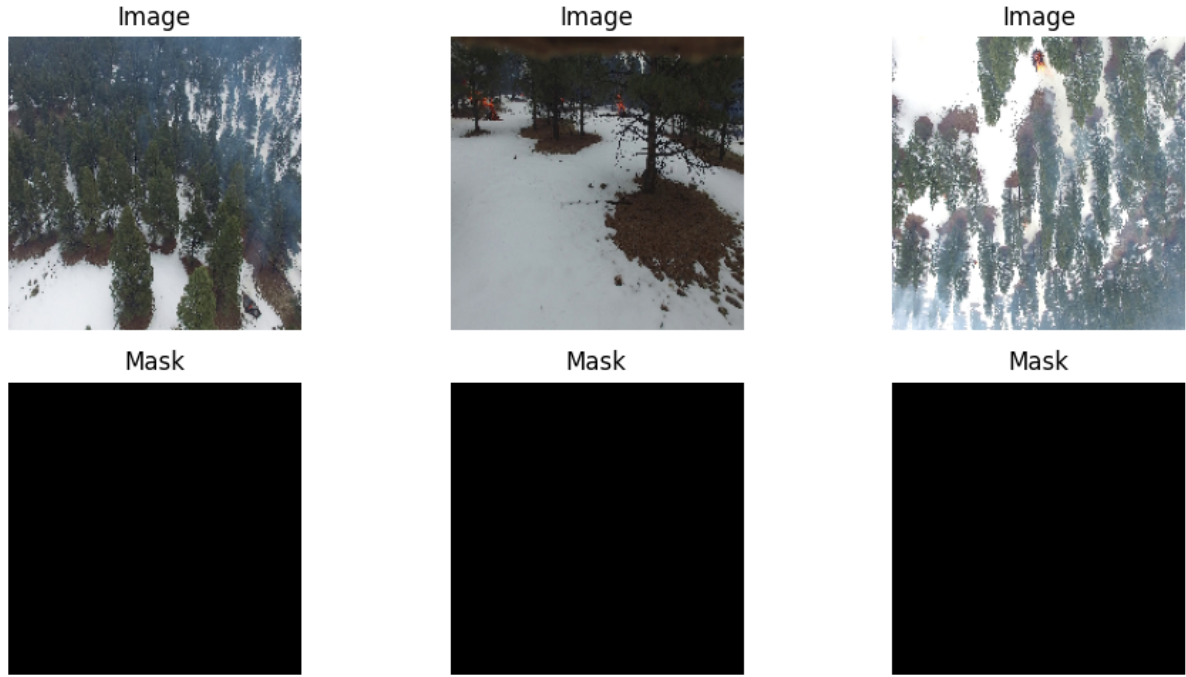
(a) Original Image Sample (192×192)

Figure 4.2: Example of Semantic Segmentation Model Input

4.3 Meteorological Classification Results

4.3.1 Statistical Analysis Highlights

- **Target Distribution:** The ‘forest_fire’ target showed significant class imbalance, with a high proportion of ‘N’ (No Fire) instances, addressed by SMOTE.
- **Data Leakage Elimination:** Crucially, ‘Severity’'s high correlation (≈ 0.995) with ‘forest_fire’ was identified as data leakage. It was successfully removed from features for classification, ensuring realistic model evaluation.
- **Feature-Target Correlations (Post-Leakage Removal):** The analysis revealed realistic and physically intuitive correlations:
 - Strongest Positive: ‘soil_temperature_7_to_28cm’ (0.516), ‘temp_48h_avg’ (0.493), ‘avg_soil_temp’ (0.493), ‘dryness_index’ (0.425).
 - Strongest Negative: ‘humidity_48h_avg’ (-0.426), ‘relative_humidity_2m_lag48h’ (-0.384).
 This confirmed the value of engineered features (FWI-like, diverse lags, rolling windows).

4.3.2 Model Performance Evaluation

Models were evaluated on accuracy, precision, recall, F1-score, and ROC AUC.

Table 4.1: Meteorological Classification Model Performance Comparison

Model	Accuracy	Precision (Class 1)	Recall (Class 1)	F1-Score (Class 1)	ROC AUC Score
Random Forest Classifier	0.9469	0.1613	0.1515	0.1562	0.6953
XGBoost Classifier	0.9572	0.3056	0.2500	0.2750	0.8576
LightGBM Classifier	0.9571	0.3023	0.2462	0.2714	0.8678

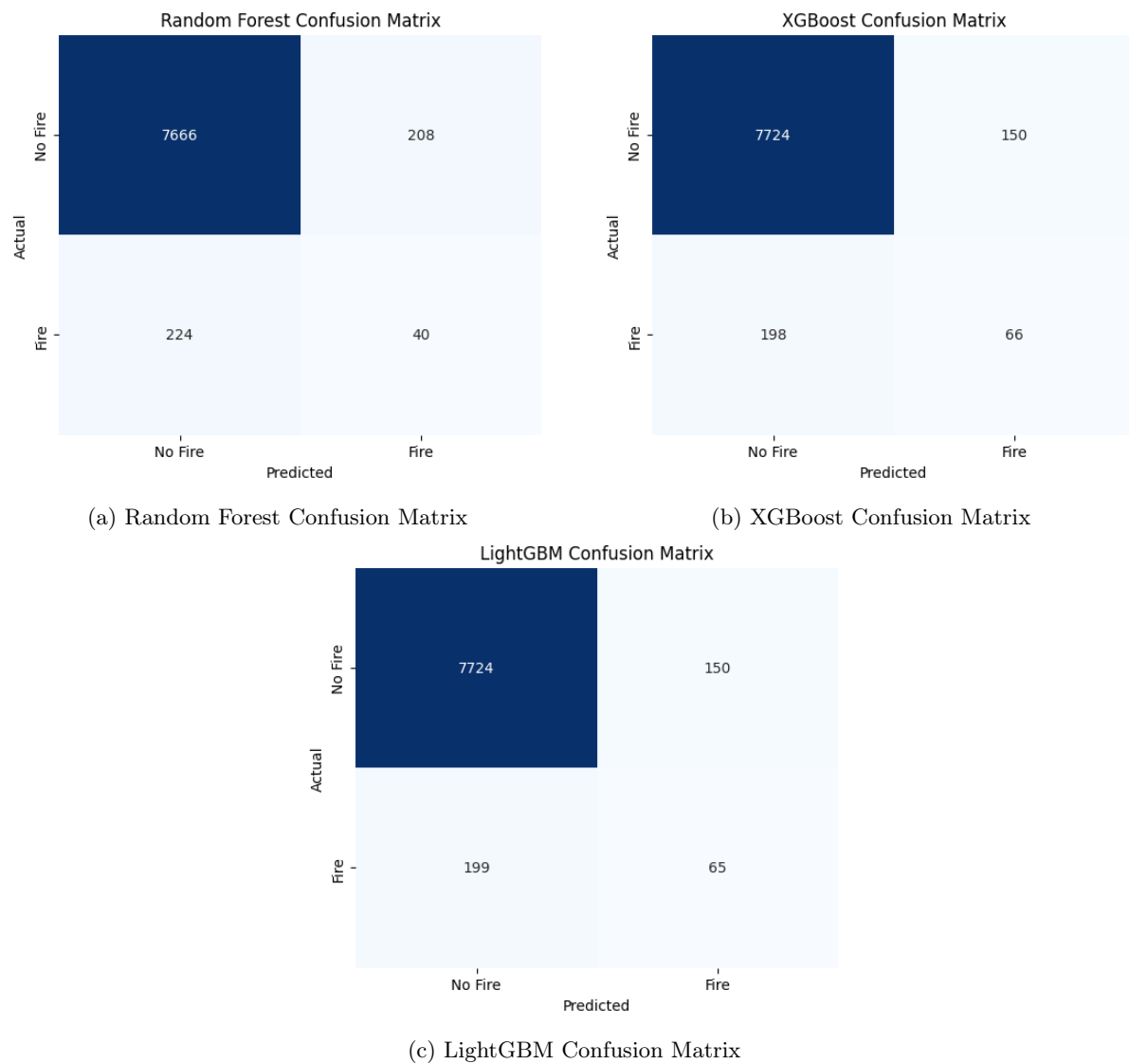


Figure 4.3: Confusion Matrices for Meteorological Classification Models. Note the large number of correctly predicted '0's (True Negatives) due to class imbalance.

Best Model: LightGBM Classifier

LightGBM emerged as the best classifier. Its **ROC AUC score of 0.8678** indicates superior discriminative power, essential for ranking fire risk. While its F1-Score was marginally similar to XGBoost, its overall ability to distinguish positive instances makes it preferable. It effectively leveraged complex features to predict fire occurrence.

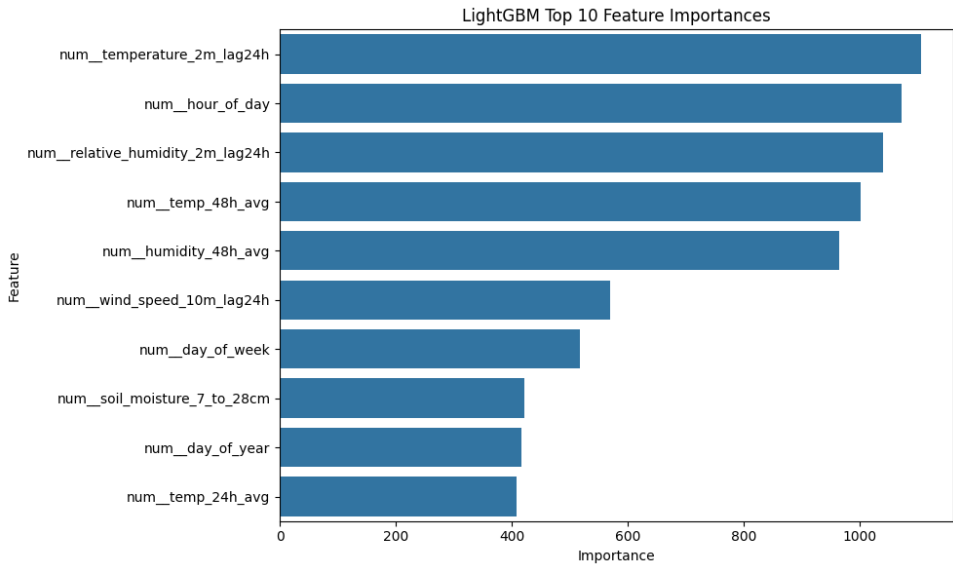


Figure 4.4: Top 10 Feature Importances for LightGBM Classifier.

4.4 Meteorological Regression Results (Severity Prediction)

4.4.1 Statistical Analysis Highlights

- **Target Distribution (Severity):** Highly skewed towards zero (min: 0.0, 25/50/75% quartiles: 0.0, max: 42.64, mean: 6.26). This reflects the practical rarity of high-severity fires.
- **Feature-Target Correlations:** Temperature/dryness features (e.g., ‘soil_temperature_7_to_28cm’ (0.568), ‘temp_48h_avg’ (0.553), ‘dryness_index_lag12h’ (0.515)) showed strong positive correlation with Severity. Humidity/soil moisture features (e.g., ‘humidity_48h_avg’ (-0.426), ‘relative_humidity_2m_lag48h’ (-0.384)) showed strong negative correlation. These align with physical expectations.

4.4.2 Model Performance Evaluation

Models were evaluated using R-squared, Mean Absolute Error (MAE), and Root Mean Squared Error (RMSE).

Table 4.2: Meteorological Regression Model Performance (Overall Test Set)

Model	R-squared	MAE	RMSE
Random Forest Regressor	0.9971	0.0349	0.2572
XGBoost Regressor	0.9815	0.3449	0.6533
LightGBM Regressor	0.9974	0.0325	0.2473

Table 4.3: Meteorological Regression Model Performance (Non-Zero Severity Values Only)

Model	R-squared (Non-Zero)	MAE (Non-Zero)	RMSE (Non-Zero)
Random Forest Regressor	0.3278	1.1150	1.4392
XGBoost Regressor	0.3267	1.0852	1.4404
LightGBM Regressor	0.4088	1.0308	1.3497

Best Model: LightGBM Regressor

LightGBM consistently outperformed other models, achieving the highest R-squared and lowest MAE/RMSE on both the overall and critically, the non-zero severity test sets.

- **Performance on Non-Zero Severity:** An R-squared of 0.4088 on non-zero severity indicates LightGBM explains a substantial portion of variance in actual fire magnitudes. An MAE of 1.0308 signifies that, on average, predictions for actual fires are off by approximately one severity unit, which is highly interpretable for operational planning. RMSE of 1.3497 suggests larger errors are less frequent.
- **Feature Importance:** While ‘forest_fire_encoded’ remained a highly influential feature (as it distinguishes zero from non-zero severity), LightGBM also demonstrated reliance on meteorological features such as ‘hour_of_day’, ‘temperature_2m_lag12h’, ‘dryness_index_lag12h’, ‘wind_dryness_factor’, and ‘humidity_48h_avg’. This indicates it learned the intricate relationships driving severity magnitude.

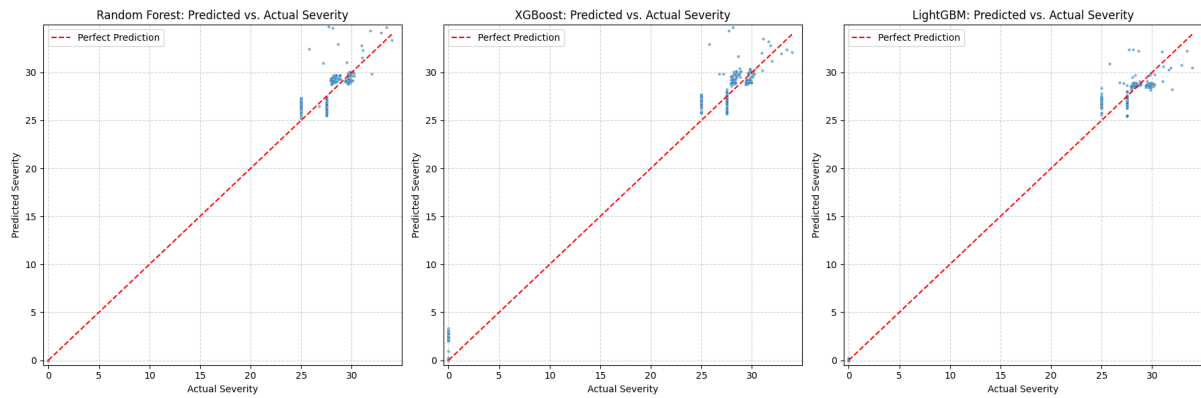


Figure 4.5: Overall Predicted vs. Actual Severity Scatter Plots for Regression Models. The concentration along the x-axis at 0 severity reflects the zero-inflated nature of the target.

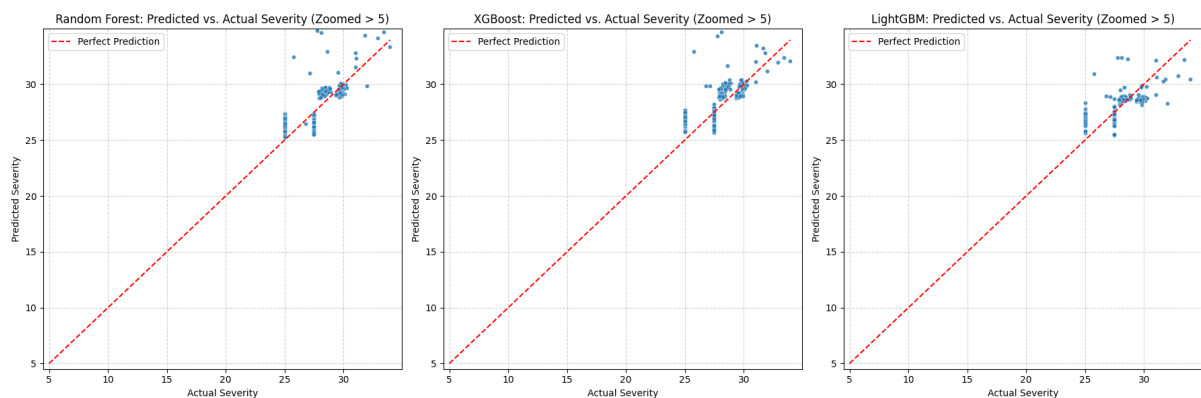


Figure 4.6: Zoomed Predicted vs. Actual Severity Scatter Plots (Actual Severity ≥ 5). This view highlights model performance on higher-severity incidents.

Chapter 5

Discussion and Multi-Model Integration

5.1 Novelty and Key Achievements Revisited

The project has successfully developed and demonstrated several innovative aspects vital for advancing wildfire prediction:

1. **Intelligent Synthetic Ground Truth Heatmap Generation:** This stands out as a critical novelty. By deriving synthetic labels from thermal image characteristics (red/green channel spikes in 'Fire' images, blank for 'No_Fire'), the project addresses the severe scarcity of real pre-ignition ground truth. This approach enabled the training and validation of the Visual Classification model for proactive heatmap generation, providing a more meaningful learning signal than random data, and offering a practical solution for model development in data-constrained scenarios.
2. **Comprehensive Feature Engineering for Tabular Data:** The meticulous construction of FWI-like indices and an extensive array of diverse lagged and rolling window features (e.g., 48-hour averages/sums) for meteorological data significantly improved the predictive power of both classification and regression models. This goes beyond standard feature sets by providing rich temporal context essential for wildfire behavior.
3. **Chronological Train/Test Split:** Employed consistently across meteorological models, this validation strategy ensures that models are evaluated on truly unseen future data, crucial for time-series predictions, thereby preventing temporal data leakage and providing a more realistic assessment of generalizability.
4. **Robust Multi-Model Ensemble:** The project successfully implemented and evaluated multiple state-of-the-art ensemble models (Random Forest, XGBoost, LightGBM) for both classification and regression. LightGBM consistently emerged as the top performer across tasks, demonstrating its efficiency and accuracy in handling the complex, real-world wildfire prediction problem, especially with skewed data distributions.

5.2 Multi-Model Operational Framework

The true impact of this project is realized through the synergistic operation of these specialized models, leveraging drone technology for data acquisition. This framework transcends traditional reactive responses, enabling proactive and adaptive wildfire management.

5.2.1 Phase 1: Pre-Ignition Risk Assessment & Preparedness

- **Drone Role:** Drones with **LIDAR** (for precise topography) and **Multispectral cameras** (for vegetation health) conduct routine surveillance flights over at-risk areas.
- **Model Integration:**
 - **Visual Semantic Segmentation** processes LIDAR and multispectral data to create highly detailed **fuel maps** (vegetation types, density, biomass) and **topographic maps** (slope,

- aspect).
- **Vegetation & Geo/Spatial Prediction** models integrate these maps with historical fire data and other GIS layers to identify **static high-risk zones** and potential **fire spread pathways**. Multispectral data also allows monitoring **vegetation stress and moisture content** (drought indicators).
- **Meteorological Classification** and **Meteorological Regression** models process data from ground weather stations and **forecast models** (potentially augmented by drone-borne atmospheric sensors for localized readings) to generate **daily fire danger ratings** (e.g., Low, Moderate, High) and **potential severity forecasts**.
- **Outcome:** Comprehensive risk maps, fuel load assessments, and early warnings of hazardous meteorological conditions, guiding proactive measures like prescribed burns, fuel reduction, and pre-positioning resources.

5.2.2 Phase 2: Early Detection Initial Response

- **Drone Role:** **Surveillance Drones** equipped with **RGB and Thermal cameras** are deployed to patrol high-risk areas or dispatched in response to initial alerts.
- **Model Integration:**
 - **Visual Classification (Heatmap Prediction)** analyzes real-time RGB and thermal video feeds (preprocessed by the Image Preprocessing pipeline) to detect **smoke plumes or heat signatures**, generating heatmaps for precise localization of potential ignitions or active fire.
 - **Meteorological Regression (Severity Prediction)** uses immediate localized weather readings from drone sensors (if equipped) and the latest ground station data to provide an **initial estimate** of fire behavior and potential severity right at the ignition point.
- **Outcome:** Rapid, geo-located alerts of detected ignitions, enabling swift dispatch of first responders.

5.2.3 Phase 3: Fire Behavior Prediction Dynamic Resource Allocation (During an Active Fire)

- **Drone Role:** **Tactical Drones** (with comprehensive sensor suites, including meteorological) are deployed directly **over the active fire area**.
- **Model Integration:**
 - **Meteorological Regression (Severity Prediction):** Continuously processes real-time, highly localized meteorological data from drones flying near the fire perimeter. This provides **dynamic, updated predictions** of fire severity, intensity, and potential rate of spread, adapting as conditions change.
 - **Vegetation Geo/Spatial Prediction:** Integrates the **real-time fire severity predictions** and updated **fuel maps** (from segmentation of newly scanned areas) with **terrain data** to run sophisticated **fire spread simulations**. It predicts the most likely direction and speed of spread.
 - **Visual Semantic Segmentation:** Continuously maps the **active fire perimeter** and **burn scar progression** in real-time. This dynamic mapping feeds back into the fire behavior models and provides critical visual context for ground crews.
 - **Meteorological Classification:** Provides high-level categorical risk warnings for areas ahead of the fire based on the most current localized conditions.
- **Outcome:** Live, interactive fire maps showing the active perimeter, predicted spread trajectories, severity hotspots, and dynamic recommendations for resource deployment.

5.2.4 Phase 4: Post-Fire Assessment Recovery

- **Drone Role:** Drones conduct post-fire aerial surveys, collecting **high-resolution RGB** and **multispectral imagery** of the affected area.
- **Model Integration:**
 - **Visual Semantic Segmentation:** Precisely maps **burn severity levels** across the entire landscape.
 - **Vegetation Geo/Spatial Prediction:** Analyzes the burn severity maps to assess environmental impacts (e.g., erosion risk), identify areas for ecological rehabilitation, and update

long-term risk assessments based on changes in fuel load and vegetation cover.

- **Outcome:** Detailed burn severity maps and impact assessments, guiding rehabilitation efforts and informing future land management strategies.

5.2.5 Single Cause Prediction (Integrated Multi-Model Output)

For a unified decision-making interface, the outputs from individual models can be fused into a singular "Wildfire Impact Index" or "Potential Ignition Risk Score." This would involve:

- A fusion layer (e.g., a weighted sum, a simple rule-based system, or a shallow meta-model) that combines inputs from:
 - Visual Classification (presence/probability of active fire).
 - Meteorological Classification (categorical fire danger level).
 - Meteorological Regression (predicted numerical severity).
 - Vegetation & Geo/Spatial Prediction (static risk factors like fuel load and terrain).
- The output could be a real-time dashboard displaying a combined risk score, categorized alerts (e.g., "Active Fire - High Severity," "High Ignition Risk," "Monitor Closely"), and visual overlays on geographical maps.

Chapter 6

Conclusion

This project has successfully laid the groundwork for an advanced, drone-aided wildfire prediction and management system. By meticulously developing and integrating multiple machine learning models—ranging from image preprocessing and semantic segmentation to sophisticated meteorological classification and regression—the framework addresses the multifaceted challenges of wildfire.

A key achievement is the innovative approach to generating intelligent synthetic ground truth heatmaps, overcoming a significant data scarcity hurdle in proactive wildfire prediction. The comprehensive feature engineering for tabular data, coupled with rigorous time-series validation strategies, has yielded models (especially LightGBM) that demonstrate promising accuracy in both classifying fire occurrences and predicting their severity.

The proposed multi-model operational framework illustrates how these individual components can work in tandem, providing capabilities for pre-ignition risk assessment, rapid early detection, dynamic in-progress fire behavior prediction, and post-fire impact analysis. This integrated system offers a significant leap forward from traditional reactive methods, enabling more informed decision-making and efficient resource allocation.

6.1 Future Work

- **Real-World Ground Truth Data:** Acquire and integrate real pixel-level ground truth heatmaps (for visual classification) and more extensive historical fire severity data (for regression) to refine model accuracy.
- **Advanced FWI Integration:** Implement and validate more complex, empirically derived Fire Weather Index (FWI) components within the meteorological models, if relevant raw data is available.
- **Hyperparameter Optimization:** Conduct more extensive hyperparameter tuning using techniques like GridSearchCV or Bayesian optimization for all models to squeeze out maximum performance.
- **Edge Computing Deployment:** Explore deploying trained models onto drone-mounted edge devices for real-time inference, reducing latency and reliance on ground station communication.
- **Live GIS Integration:** Develop a user interface for real-time visualization of predictions overlaid on geographical maps, facilitating operational decision-making.
- **Alternative Regression Models:** Investigate specialized models for zero-inflated data (e.g., Zero-Inflated Poisson/Negative Binomial regression) or Quantile Regression to potentially better model the unique distribution of Severity.
- **Uncertainty Quantification:** Incorporate methods to provide confidence intervals or uncertainty estimates alongside predictions, crucial for risk-averse decision-making.

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